

CLASS 10 · ICSE · SAMPLE PAPER SOLUTIONS

MathSetu Sample Paper 1: Mathematics

Step-by-step solutions with the marks for each question

The question numbers match the question paper.

1 Section A, Question 1: choose the correct answer (15 marks)

1. An article is marked at ₹800 and sold within the same state. If the rate of GST is 18%, the CGST charged is: (a) ₹36 (b) ₹72 (c) ₹80 (d) ₹144 **1 mark**

SOLUTION

For a sale within a state, CGST is half the GST rate: 9% of ₹800 = ₹72.

Answer: (b) ₹72

2. ₹500 is deposited every month for 12 months in a recurring deposit account. The equivalent principal for one month, used to calculate the interest, is: (a) ₹6000 (b) ₹33,000 (c) ₹39,000 (d) ₹78,000 **1 mark**

SOLUTION

Equivalent principal for one month = $P \times \frac{n(n+1)}{2} = 500 \times \frac{12 \times 13}{2} = ₹39,000$.

Answer: (c) ₹39,000

3. A ₹100 share is bought at ₹125 and pays a dividend of 10%. The rate of return on the investment is: (a) 8% (b) 10% (c) 12.5% (d) 25% **1 mark**

SOLUTION

Dividend on one share = ₹10 on an investment of ₹125, so the return is $\frac{10}{125} \times 100 = 8\%$.

Answer: (a) 8%

4. The solution set of $-3 < 2x - 1 \leq 5$, where x is a whole number, is: (a) $\{1, 2, 3\}$ (b) $\{0, 1, 2\}$ (c) $\{0, 1, 2, 3\}$ (d) $\{-1, 0, 1, 2, 3\}$ **1 mark**

SOLUTION

Add 1 throughout: $-2 < 2x \leq 6$, so $-1 < x \leq 3$. The whole numbers in this range are 0, 1, 2, 3.

Answer: (c) $\{0, 1, 2, 3\}$

5. The roots of $x^2 - 6x + 9 = 0$ are: (a) real and unequal (b) real and equal (c) not real (d) equal in size and opposite in sign **1 mark**

SOLUTION

$D = (-6)^2 - 4(1)(9) = 0$, so the roots are real and equal (both are 3).

Answer: (b) real and equal

6. The mean proportional between 4 and 25 is: (a) 10 (b) 14.5 (c) 21 (d) 100 **1 mark**

SOLUTION

Mean proportional = $\sqrt{4 \times 25} = 10$.

Answer: (a) 10

7. The remainder when $x^3 - 2x^2 + 3x - 4$ is divided by $x - 2$ is: (a) -4 (b) 0 (c) 2 (d) 6 **1 mark**

SOLUTION

By the Remainder Theorem, the remainder is $f(2) = 8 - 8 + 6 - 4 = 2$.

Answer: (c) 2

8. A is a matrix of order 2×3 and B is a matrix of order 3×1 . The order of AB is: (a) 3×3 (b) 2×1 (c) 1×2 (d) AB is not possible **1 mark**

SOLUTION

The columns of A (3) equal the rows of B (3), so AB exists, with the rows of A and the columns of B : 2×1 .

Answer: (b) 2×1

9. The n th term of an A.P. is $5n - 2$. Its common difference is: (a) -2 (b) 2 (c) 3 (d) 5 **1 mark**

SOLUTION

$T_1 = 3$ and $T_2 = 8$, so $d = 8 - 3 = 5$.

Answer: (d) 5

10. The 5th term of the G.P. 3, 6, 12, ... is: (a) 24 (b) 36 (c) 48 (d) 96 **1 mark**

SOLUTION

$a = 3, r = 2: T_5 = ar^4 = 3 \times 16 = 48$.

Answer: (c) 48

11. The image of the point $(3, -4)$ after reflection in the y -axis is: (a) $(3, 4)$ (b) $(-3, 4)$ (c) $(-3, -4)$ (d) $(-4, 3)$ **1 mark**

SOLUTION

Reflection in the y -axis changes the sign of the x -coordinate only: $(3, -4) \rightarrow (-3, -4)$.

Answer: (c) $(-3, -4)$

12. The slope of the line $3x - 4y + 7 = 0$ is: (a) $\frac{3}{4}$ (b) $-\frac{3}{4}$ (c) $\frac{4}{3}$ (d) $\frac{7}{4}$ **1 mark**

SOLUTION

Write it as $y = \frac{3}{4}x + \frac{7}{4}$. The slope is $\frac{3}{4}$.

Answer: (a) $\frac{3}{4}$

13. An arc of a circle subtends an angle of 110° at the centre. The angle it subtends at a point on the remaining part of the circle is: (a) 55° (b) 70° (c) 110° (d) 220° **1 mark**

SOLUTION

The angle at the centre is twice the angle at the circumference: $\frac{110^\circ}{2} = 55^\circ$.

Answer: (a) 55°

14. The volume of a cone of radius 3 cm and height 7 cm is: (a) 22 cm^3 (b) 66 cm^3 (c) 132 cm^3 (d) 198 cm^3 **1 mark**

SOLUTION

Volume = $\frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 9 \times 7 = 66 \text{ cm}^3$.

Answer: (b) 66 cm^3

15. A letter is chosen at random from the letters of the word TRIANGLE. The probability that it is a vowel is: (a) $\frac{1}{4}$ (b) $\frac{3}{8}$ (c) $\frac{1}{2}$ (d) $\frac{5}{8}$ **1 mark**

SOLUTION

The word has 8 letters, of which 3 (I, A, E) are vowels. Probability = $\frac{3}{8}$.

Answer: (b) $\frac{3}{8}$

2 Section A, Question 2 (12 marks)

16. Meera deposits ₹1200 every month for 2 years in a recurring deposit account that pays simple interest at 8% per annum. Find the interest she earns and the amount she gets on maturity. **4 marks**

SOLUTION

$$P = ₹1200, n = 24 \text{ months}, r = 8\%.$$

$$\text{Interest} = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100} = 1200 \times \frac{24 \times 25}{24} \times \frac{8}{100} = ₹2400.$$

$$\text{Total deposited} = 1200 \times 24 = ₹28,800, \text{ so the maturity amount} = 28,800 + 2400 = ₹31,200.$$

Answer: interest ₹2400, maturity amount ₹31,200

17. If $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$, find $A^2 - 2B$. **4 marks**

SOLUTION

$$A^2 = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 4-1 & 2+3 \\ -2+3 & -1+9 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix}$$

$$2B = \begin{bmatrix} 2 & 0 \\ 4 & -2 \end{bmatrix}$$

$$A^2 - 2B = \begin{bmatrix} 3-2 & 5-0 \\ -5-4 & 8+2 \end{bmatrix}$$

$$\text{Answer: } \begin{bmatrix} 1 & 5 \\ -9 & 10 \end{bmatrix}$$

18. Prove that $\frac{\cos A}{1 - \sin A} + \frac{\cos A}{1 + \sin A} = 2 \sec A$. **4 marks**

SOLUTION

$$\text{LHS} = \frac{\cos A (1 + \sin A) + \cos A (1 - \sin A)}{(1 - \sin A)(1 + \sin A)}$$

$$= \frac{2 \cos A}{1 - \sin^2 A} = \frac{2 \cos A}{\cos^2 A}$$

$$= \frac{2}{\cos A} = 2 \sec A = \text{RHS. Proved.}$$

3 Section A, Question 3 (13 marks)

19. A solid metal sphere of radius 6 cm is melted and recast into a solid cylinder of radius 4 cm. Find the height of the cylinder and its curved surface area, correct to one decimal place. **4 marks**

SOLUTION

$$\text{Volume of the sphere} = \frac{4}{3} \pi \times 6^3 = 288\pi \text{ cm}^3.$$

$$\text{Volume of the cylinder} = \pi \times 4^2 \times h = 16\pi h. \text{ So } 16\pi h = 288\pi \text{ and } h = 18 \text{ cm.}$$

$$\text{Curved surface area} = 2\pi rh = 2 \times \frac{22}{7} \times 4 \times 18 = \frac{3168}{7} \approx 452.6 \text{ cm}^2.$$

Answer: height 18 cm, curved surface area $\approx 452.6 \text{ cm}^2$

- 20.** $A(2, 3)$ and $B(6, -5)$ are two points. Find (i) the slope of AB (ii) the equation of the perpendicular bisector of AB (iii) the point where this bisector meets the x -axis. **4 marks**

SOLUTION

(i) Slope of $AB = \frac{-5 - 3}{6 - 2} = -2$.

(ii) The mid-point of AB is $\left(\frac{2+6}{2}, \frac{3-5}{2}\right) = (4, -1)$, and the slope of the perpendicular bisector is $\frac{1}{2}$.

Its equation is $y + 1 = \frac{1}{2}(x - 4)$, that is $x - 2y - 6 = 0$.

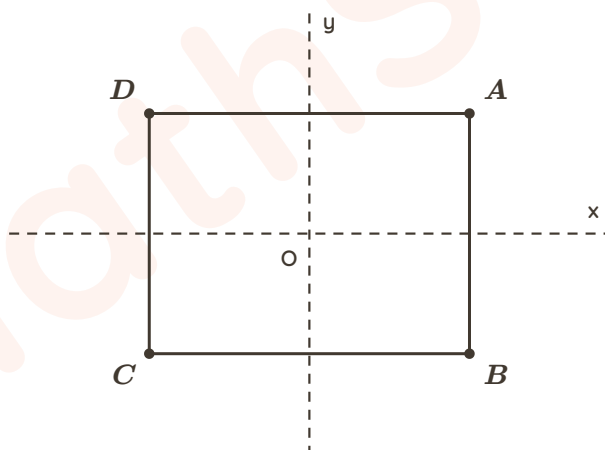
(iii) Put $y = 0$: $x = 6$. The point is $(6, 0)$.

Answer: (i) -2 (ii) $x - 2y - 6 = 0$ (iii) $(6, 0)$

- 21.** Use graph paper for this question, with $1 \text{ cm} = 1 \text{ unit}$ on both axes. Plot the point $A(4, 3)$. Reflect A in the x -axis to get B , reflect A in the origin to get C , and reflect A in the y -axis to get D . (i) Write the coordinates of B , C and D . (ii) Name the figure $ABCD$. (iii) Find its perimeter and its area. **5 marks**

SOLUTION

(i) $B = (4, -3)$, $C = (-4, -3)$ and $D = (-4, 3)$.



(ii) AB and CD are vertical, BC and DA are horizontal, and all the angles are right angles, so $ABCD$ is a rectangle.

(iii) $AB = 6$ units and $BC = 8$ units. Perimeter $= 2(6 + 8) = 28$ units and area $= 6 \times 8 = 48$ square units.

Answer: (i) $B(4, -3)$, $C(-4, -3)$, $D(-4, 3)$ (ii) a rectangle (iii) perimeter 28 units, area 48 square units

4 Section B, Question 4 (10 marks)

- 22.** A dealer in Delhi sells a washing machine, marked at ₹20,000, to a customer in Delhi at a discount of 10%. The rate of GST is 18%. Find the CGST, the SGST and the total amount the customer pays. **3 marks**

SOLUTION

Selling price after discount = 20,000 – 10% of 20,000 = ₹18,000.

CGST = 9% of 18,000 = ₹1620 and SGST = 9% of 18,000 = ₹1620.

Amount paid = 18,000 + 1620 + 1620 = ₹21,240.

Answer: CGST ₹1620, SGST ₹1620, amount paid ₹21,240

- 23.** Solve $x^2 - 5x - 3 = 0$, giving the answers correct to two decimal places. **3 marks**

SOLUTION

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{5 \pm \sqrt{25 + 12}}{2} = \frac{5 \pm \sqrt{37}}{2}, \text{ and } \sqrt{37} \approx 6.083.$$

$$x = \frac{5 + 6.083}{2} \approx 5.54 \text{ or } x = \frac{5 - 6.083}{2} \approx -0.54.$$

Answer: $x = 5.54$ or $x = -0.54$

- 24.** Find the mean of the following distribution by the step-deviation method. **4 marks**

Class	0 - 20	20 - 40	40 - 60	60 - 80	80 - 100
Frequency	6	10	16	12	6

SOLUTION

Take the assumed mean $A = 50$ and the class width $i = 20$, with $t = \frac{x - A}{i}$.

Class	f	x	t	ft
0 - 20	6	10	-2	-12
20 - 40	10	30	-1	-10
40 - 60	16	50	0	0
60 - 80	12	70	1	12
80 - 100	6	90	2	12
Total	50			2

$$\text{Mean} = A + \frac{\sum ft}{\sum f} \times i = 50 + \frac{2}{50} \times 20 = 50.8.$$

Answer: mean = 50.8

5 Section B, Question 5 (10 marks)

25. A man invests ₹9600 in ₹100 shares at a premium of 20%. The company declares a dividend of 15%. Find (i) the number of shares he buys (ii) his annual dividend (iii) the rate of return on his investment.

3 marks

SOLUTION

(i) Market value of one share = $100 + 20 = ₹120$. Number of shares = $\frac{9600}{120} = 80$.

(ii) Dividend on one share = 15% of ₹100 = ₹15. Annual dividend = $80 \times 15 = ₹1200$.

(iii) Rate of return = $\frac{1200}{9600} \times 100 = 12.5\%$.

Answer: (i) 80 shares (ii) ₹1200 (iii) 12.5%

26. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 4)^\circ$, $\angle B = (x + 40)^\circ$ and $\angle C = (3x + 16)^\circ$. Find x and all four angles. 3 marks

SOLUTION

Opposite angles of a cyclic quadrilateral add up to 180° : $(2x + 4) + (3x + 16) = 180$, so $5x = 160$ and $x = 32$.

$\angle A = 68^\circ$, $\angle C = 112^\circ$, $\angle B = 72^\circ$ and $\angle D = 180^\circ - 72^\circ = 108^\circ$.

Answer: $x = 32$; the angles are 68° , 72° , 112° and 108°

27. Use the Factor Theorem to show that $x - 2$ is a factor of $2x^3 + x^2 - 13x + 6$. Hence factorise the expression completely. 4 marks

SOLUTION

Let $f(x) = 2x^3 + x^2 - 13x + 6$. Then $f(2) = 16 + 4 - 26 + 6 = 0$, so $x - 2$ is a factor.

Dividing $f(x)$ by $x - 2$ gives the quotient $2x^2 + 5x - 3$.

$2x^2 + 5x - 3 = 2x^2 + 6x - x - 3 = 2x(x + 3) - 1(x + 3) = (2x - 1)(x + 3)$.

Answer: $(x - 2)(2x - 1)(x + 3)$

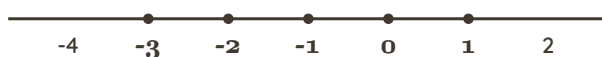
6 Section B, Question 6 (10 marks)

28. Solve $3x - 7 \leq 7x + 5 < 19$, where x is an integer, and show the solution set on a number line. 3 marks

SOLUTION

From $3x - 7 \leq 7x + 5$: $-12 \leq 4x$, so $x \geq -3$. From $7x + 5 < 19$: $7x < 14$, so $x < 2$.

So $-3 \leq x < 2$, and the solution set is $\{-3, -2, -1, 0, 1\}$. On the number line these five integers are marked with dots.



Answer: $\{-3, -2, -1, 0, 1\}$

- 29.** The 4th term of an A.P. is 22 and its 15th term is 66. Find the first term, the common difference and the sum of its first 8 terms. **3 marks**

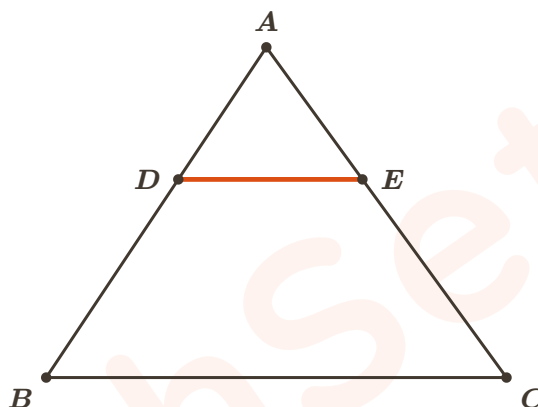
SOLUTION

$a + 3d = 22$ and $a + 14d = 66$. Subtracting, $11d = 44$, so $d = 4$ and $a = 10$.

$$S_8 = \frac{8}{2} [2 \times 10 + 7 \times 4] = 4 \times 48 = 192.$$

Answer: $a = 10, d = 4, S_8 = 192$

- 30.** In $\triangle ABC$, D lies on AB and E lies on AC so that $DE \parallel BC$ and $AD : DB = 2 : 3$. (i) Prove that $\triangle ADE \sim \triangle ABC$. (ii) If $DE = 4$ cm, find BC . (iii) Find the ratio of the area of $\triangle ADE$ to the area of trapezium $DBCE$. **4 marks**



SOLUTION

(i) $\angle A$ is common and $\angle ADE = \angle ABC$ (corresponding angles, $DE \parallel BC$). So $\triangle ADE \sim \triangle ABC$ (AA).

(ii) $\frac{AD}{AB} = \frac{2}{2+3} = \frac{2}{5}$, so $\frac{DE}{BC} = \frac{2}{5}$ and $BC = \frac{5}{2} \times 4 = 10$ cm.

(iii) $\frac{\text{area of } \triangle ADE}{\text{area of } \triangle ABC} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$. So the trapezium has $25 - 4 = 21$ of the 25 parts.

Answer: (ii) $BC = 10$ cm (iii) $4 : 21$

7 Section B, Question 7 (10 marks)

- 31.** What number must be added to each of 2, 6, 8 and 18 so that the four results are in proportion? **3 marks**

SOLUTION

Let the number be x . Then $\frac{2+x}{6+x} = \frac{8+x}{18+x}$.

$(2+x)(18+x) = (6+x)(8+x)$ gives $36 + 20x + x^2 = 48 + 14x + x^2$, so $6x = 12$ and $x = 2$.

Check: 4, 8, 10, 20 are in proportion, since $4 \times 20 = 8 \times 10$.

Answer: 2

- 32.** The 3rd term of a G.P. is 12 and its 6th term is 96. Find the first term, the common ratio and the sum of its first 8 terms. **3 marks**

SOLUTION

$ar^2 = 12$ and $ar^5 = 96$. Dividing, $r^3 = 8$, so $r = 2$ and $a = 3$.

$$S_8 = \frac{a(r^8 - 1)}{r - 1} = \frac{3(256 - 1)}{2 - 1} = 765.$$

Answer: $a = 3$, $r = 2$, $S_8 = 765$

- 33.** From a point A on level ground, the angle of elevation of the top of a tower is 30° . On walking 40 m towards the tower to a point B , the angle of elevation becomes 60° . Find the height of the tower, correct to two decimal places, and the distance of B from the tower. (Take $\sqrt{3} = 1.732$.) **4 marks**

SOLUTION

Let the tower be h m high and let B be x m from its foot.

From B : $\tan 60^\circ = \frac{h}{x}$, so $h = \sqrt{3}x$. From A : $\tan 30^\circ = \frac{h}{x + 40}$, so $x + 40 = \sqrt{3}h$.

Then $x + 40 = \sqrt{3} \times \sqrt{3}x = 3x$, so $x = 20$ m.

$h = 20\sqrt{3} = 20 \times 1.732 = 34.64$ m.

Answer: height 34.64 m; B is 20 m from the tower

8 Section B, Question 8 (10 marks)

- 34.** Cards numbered 1 to 30 are put in a box and one card is drawn at random. Find the probability that the number on it is (i) a multiple of 4 (ii) a prime number (iii) a perfect square. **3 marks**

SOLUTION

(i) Multiples of 4: 4, 8, 12, 16, 20, 24, 28, which is 7 cards. Probability = $\frac{7}{30}$.

(ii) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, which is 10 cards. Probability = $\frac{10}{30} = \frac{1}{3}$.

(iii) Perfect squares: 1, 4, 9, 16, 25, which is 5 cards. Probability = $\frac{5}{30} = \frac{1}{6}$.

Answer: (i) $\frac{7}{30}$ (ii) $\frac{1}{3}$ (iii) $\frac{1}{6}$

- 35.** In what ratio does the y -axis divide the line segment joining $A(-4, 1)$ and $B(6, 6)$? Also find the point of division. **3 marks**

SOLUTION

Let the ratio be $k : 1$. The x -coordinate of the point of division is 0: $\frac{6k - 4}{k + 1} = 0$, so $k = \frac{2}{3}$. The ratio is 2 : 3.

$$y = \frac{2 \times 6 + 3 \times 1}{2 + 3} = \frac{15}{5} = 3.$$

Answer: ratio 2 : 3, point (0, 3)

- 36.** A conical tent is 12 m high and the radius of its base is 9 m. Find (i) the cost of the canvas for its curved surface at ₹70 per m^2 (ii) the volume of air inside it, correct to the nearest m^3 . **4 marks**

SOLUTION

Slant height $l = \sqrt{9^2 + 12^2} = 15$ m.

(i) Curved surface area $= \pi r l = \frac{22}{7} \times 9 \times 15 = \frac{2970}{7} \text{ m}^2$. Cost $= \frac{2970}{7} \times 70 = ₹29,700$.

(ii) Volume $= \frac{1}{3} \pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 81 \times 12 = \frac{7128}{7} \approx 1018 \text{ m}^3$.

Answer: (i) ₹29,700 (ii) about 1018 m^3

9 Section B, Question 9 (10 marks)

- 37.** The length of a rectangular garden is 4 m more than its breadth, and its area is 96 m^2 . Find its length, its breadth and its perimeter. **3 marks**

SOLUTION

Let the breadth be x m. Then $x(x + 4) = 96$, so $x^2 + 4x - 96 = 0$.

$(x + 12)(x - 8) = 0$, so $x = 8$ (a length cannot be negative).

Breadth = 8 m, length = 12 m, perimeter = $2(8 + 12) = 40$ m.

Answer: length 12 m, breadth 8 m, perimeter 40 m

- 38.** (i) Chords AB and CD of a circle meet at a point P inside the circle. If $AP = 6$ cm, $PB = 4$ cm and $CP = 3$ cm, find CD . (ii) From a point T outside another circle, a tangent TQ and a secant TRS are drawn, with $TR = 4$ cm and $RS = 5$ cm. Find TQ . **3 marks**

SOLUTION

(i) $AP \times PB = CP \times PD$, so $6 \times 4 = 3 \times PD$ and $PD = 8$ cm. Then $CD = 3 + 8 = 11$ cm.

(ii) $TQ^2 = TR \times TS = 4 \times (4 + 5) = 36$, so $TQ = 6$ cm.

Answer: (i) $CD = 11$ cm (ii) $TQ = 6$ cm

- 39.** The marks of 11 students in a test are 12, 18, 25, 9, 30, 21, 15, 27, 33, 6 and 24. Find (i) the median (ii) the lower quartile (iii) the upper quartile (iv) the interquartile range. **4 marks**

SOLUTION

In ascending order: 6, 9, 12, 15, 18, 21, 24, 25, 27, 30, 33. Here $n = 11$.

(i) Median $= \left(\frac{n+1}{2}\right)$ th term = 6th term = 21.

(ii) Lower quartile $Q_1 = \left(\frac{n+1}{4}\right)$ th term = 3rd term = 12.

(iii) Upper quartile $Q_3 = \left(\frac{3(n+1)}{4}\right)$ th term = 9th term = 27.

(iv) Interquartile range $= Q_3 - Q_1 = 27 - 12 = 15$.

Answer: (i) 21 (ii) 12 (iii) 27 (iv) 15

10 Section B, Question 10 (10 marks)

- 40.** Find the equation of the line that passes through $(2, -1)$ and is parallel to the line $3x + 4y = 7$. Also write its y -intercept. **3 marks**

SOLUTION

The slope of $3x + 4y = 7$ is $-\frac{3}{4}$, and a parallel line has the same slope.

$y + 1 = -\frac{3}{4}(x - 2)$ gives $4y + 4 = -3x + 6$, that is $3x + 4y - 2 = 0$.

Put $x = 0$: $y = \frac{1}{2}$.

Answer: $3x + 4y - 2 = 0$, with y -intercept $\frac{1}{2}$

- 41.** Find x and y if $\begin{bmatrix} 2 & x \\ y & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$. **3 marks**

SOLUTION

Multiplying, $\begin{bmatrix} 2 + 2x \\ y + 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$.

So $2 + 2x = 8$ gives $x = 3$, and $y + 6 = 7$ gives $y = 1$.

Answer: $x = 3, y = 1$

- 42.** Using a ruler and compasses only, construct $\triangle ABC$ with $AB = 5$ cm, $BC = 6$ cm and $\angle ABC = 60^\circ$. Construct the incircle of the triangle and measure its radius. **4 marks**

SOLUTION

Draw $BC = 6$ cm. At B construct an angle of 60° and cut off $BA = 5$ cm on its arm. Join AC .

Construct the bisectors of $\angle B$ and $\angle C$. They meet at the incentre I .

From I draw IN perpendicular to BC , meeting it at N . With centre I and radius IN draw the circle. It touches all three sides.

On measuring, IN comes to about 1.6 cm.

Answer: the radius of the incircle measures about 1.6 cm

This is a practice paper written by MathSetu on the pattern of the board examination. It is not an official paper of any board. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.